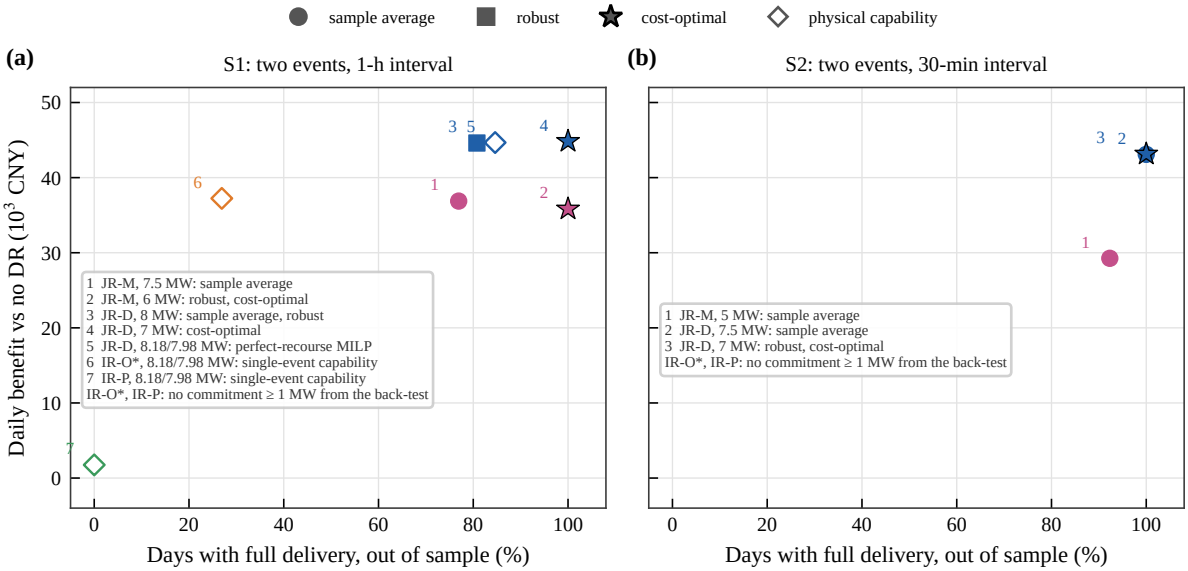
sample average
robust
cost-optimal
physical capability
(a)
S1: two events, 1-h interval
Daily benefit vs no DR (10^3 CNY)
Days with full delivery, out of sample (%)
1 JR-M, 7.5 MW: sample average
2 JR-M, 6 MW: robust, cost-optimal
3 JR-D, 8 MW: sample average, robust
4 JR-D, 7 MW: cost-optimal
5 JR-D, 8.18/7.98 MW: perfect-recourse MILP
6 IR-O*, 8.18/7.98 MW: single-event capability
7 IR-P, 8.18/7.98 MW: single-event capability
IR-O*, IR-P: no commitment ≥ 1 MW from the back-test
(b)
S2: two events, 30-min interval
Days with full delivery, out of sample (%)
1 JR-M, 5 MW: sample average
2 JR-D, 7.5 MW: sample average
3 JR-D, 7 MW: robust, cost-optimal
IR-O*, IR-P: no commitment ≥ 1 MW from the back-test